CS230: Course Outline and Syllabus

Recommended reference book: A. Tanenbaum and M. Van Steen, "Distributed Systems: Principles and Paradigms," Second Edition.

I. Introduction
   i. Historical Perspective
   ii. The need for Concurrency
   iii. Definition of Distributed Systems
   iv. Architectures and Classification
II. Use of Concurrency
   i. Speedup with Multiple Computational Resources
   ii. The Importance of the Interconnection Network
   iii. Examples – Matrix Multiplication
   iv. 2D FFT
III. Communications and the OSI model
   i. Different Network Topologies
   ii. Point-to-point vs. switching
   iii. Multi-Stage Interconnection Networks
   iv. The OSI model
IV. Operating Systems
   i. Job Tracking, Context switching.
   ii. Functions:
      - Scheduling
      - Load Balancing
   iii. Fault Tolerance and Reliability
      - Fault Classification
      - Redundancy
      - Consensus
      - Checkpointing
      - ABFT
   iv. Examples
V. Other applications of Distributed Systems
   i. Data Bases
   ii. Transaction Processing
   iii. The Internet !!
VI. Take Home Exam
VII. Research Term Papers

Final Grade= (Take Home Exam + Term Paper)/2